

II Vertiefung Integral

1 Ober- und Untersumme

Beispiel II.1 (Fläche unter dem Graphen)

$$f(x) = x^2$$

Bestimme die Fläche zwischen G_f und der x-Achse im Intervall $I = [0, 2]$.

Lösung

- in 12:

$$\begin{aligned} A &= \int_0^2 f(x) \, dx = \left[F(x) \right]_0^2 \\ &= \left[\frac{1}{3}x^3 \right]_0^2 = \frac{1}{3} \cdot 2^3 - \frac{1}{3} \cdot 0^3 = \frac{8}{3} \end{aligned}$$

- in 13:

1.1 Aufgaben

Übung II.1 S. 53/2

2. a) $f(x) = -\frac{1}{4}x^2 - x + 4$; $I = [-6; -2]$; $n = 5$

- Ableitung und Monotonie:

$$f'(x) = -\frac{1}{2}x - 1$$

$$f'(x) = 0$$

$$-\frac{1}{2}x - 1 = 0 \quad | \quad +1$$

$$-\frac{1}{2}x = 1 \quad | \quad \cdot (-2)$$

$$x_1 = -2 \quad (e)$$

	$-\infty < x < -2$	$x = -2$	$-2 < x < \infty$
$f'(x)$	+	0	-
G_f	$\nearrow$	HoP	$\searrow$

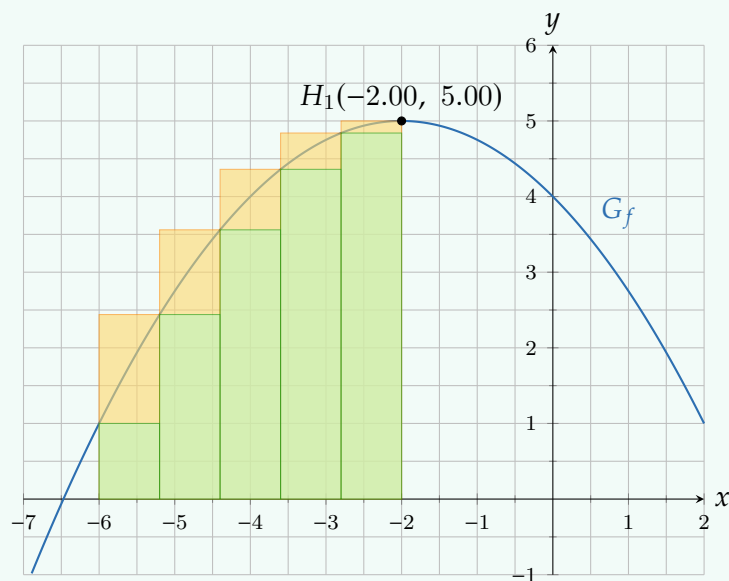
$\Rightarrow G_f$ ist sms in $] -\infty; -2]$ und smf in $[-2; \infty [$

- Ober- und Untersumme:

$$\begin{aligned} O &= \frac{4}{5} \cdot \left[f\left(-\frac{26}{5}\right) + f\left(-\frac{22}{5}\right) + f\left(-\frac{18}{5}\right) + f\left(-\frac{14}{5}\right) + f\left(-\frac{10}{5}\right) \right] \\ &= \frac{4}{5} \cdot \left[\frac{61}{25} + \frac{89}{25} + \frac{109}{25} + \frac{121}{25} + \frac{125}{25} \right] \\ &= \frac{404}{25} = 16,16 \end{aligned}$$

$$\begin{aligned} U &= O - \frac{4}{5} \cdot f\left(-\frac{10}{5}\right) + \frac{4}{5} \cdot f\left(-\frac{30}{5}\right) \\ &= \frac{404}{25} - \frac{4}{5} \cdot \frac{125}{25} + \frac{4}{5} \cdot \frac{25}{25} \\ &= \frac{324}{25} = 12,96 \end{aligned}$$

- Zeichnung:



b) $f(x) = \frac{1}{3}x^2 - x - 3; \quad I = [2; 4]; \quad n = 4$

- Ableitung und Monotonie:

$$f'(x) = \frac{2}{3}x - 1$$

$$f'(x) = 0$$

$$\frac{2}{3}x - 1 = 0 \quad | \quad +1$$

$$\frac{2}{3}x = 1 \quad | \quad \cdot \frac{3}{2}$$

$$x_1 = \frac{3}{2} \quad (e)$$

	$-\infty < x < \frac{3}{2}$	$x = \frac{3}{2}$	$\frac{3}{2} < x < \infty$
$f'(x)$	-	0	+
G_f	$\searrow$	TiP	$\nearrow$

$\Rightarrow G_f$ ist smf in $]-\infty; \frac{3}{2}]$ und sms in $[\frac{3}{2}; \infty[$

- Ober- und Untersumme:

$$\begin{aligned} O &= \frac{2}{4} \cdot \left[f\left(\frac{8}{4}\right) + f\left(\frac{10}{4}\right) + f\left(\frac{12}{4}\right) + f\left(\frac{14}{4}\right) \right] \\ &= \frac{2}{4} \cdot \left[\left| -\frac{11}{3} \right| + \left| -\frac{41}{12} \right| + |-3| + \left| -\frac{29}{12} \right| \right] \\ &= \frac{25}{4} = 6,25 \end{aligned}$$

$$\begin{aligned} U &= O - \frac{2}{4} \cdot f\left(\frac{8}{4}\right) + \frac{2}{4} \cdot f\left(\frac{16}{4}\right) \\ &= \frac{25}{4} - \frac{2}{4} \cdot \left| -\frac{11}{3} \right| + \frac{2}{4} \cdot \left| -\frac{5}{3} \right| \\ &= \frac{21}{4} = 5,25 \end{aligned}$$

- Zeichnung:

